# Documentary

## Business Problem Statement

Kraven is a mid-sized, multi-region consumer goods company operating across international markets. The company sells a diversified product portfolio across multiple customer segments, using promotional discounting to drive volume.

Between 2013 and 2014, Kraven experienced steady revenue growth, but leadership has growing concerns about margin erosion, discount effectiveness, and uneven profitability across products and customer segments. Current reporting provides high-level totals but lacks clarity around profit drivers and future performance expectations.

Senior leadership requires a detailed financial analysis built entirely in Excel to support budgeting, pricing strategy, and operational planning.

## Data Collection, Understanding & Validation

Dataset: https://learn.microsoft.com/en-us/power-bi/create-reports/sample-financial-download
Fields/ Columns: 16
Rows: 700
Date Range: 2013 - 2014
Blank Rows: No

## Dataset Overview

| Columns | Descriptions |
|---|---|
| Segment | Customer segments (Government, Midmarket, Channel Partners and Small Business) |
| Country | List of countries (Germany, France, Canada, Mexico and USA) |
| Product | Products available for sales (Amarilla, VTT, Paseo, Montana, Velo and Carratera) |
| Discount Band | Discount bins (High, Medium, Low and None) |
| Unit Sold | Quantity of goods sold |
| Manufacturing Price | The total cost incurred to produce a product |
| Sales Price | The selling price of a product to the customers |
| Gross Sales | The total amount received for a particular sale (Units Sold * Sales Price) |
| Discounts | The total amount of discount given to customers |

| | |
|---|---|
| Sales | The net sales after deducting discounts (Gross Sales - Discounts) |
| COGS | The direct cost incurred to produce or acquire a goods |
| Profit | The Gross profits made after deducting COGS (Sales - COGS) |
| Date | Different dates of each transaction |
| Month Number | Month Number (1-12) |
| Month Name | Month Name (January - December) |
| Year | Year (2013 - 2014) |

## Data Cleaning & Preparations

Data preparation was performed using **Excel Power Query** to ensure analytical readiness and consistency across tables.

Key steps included:

- Adding new fields from existing fields (e.g., *Id, Month Name[short], Profit Margin*) for more context.
- Recalculate and validate all core financial metrics (e.g., Gross Sales, Net Sales and Profit)
- Eliminated duplicate records to maintain data integrity.
- Handled missing values through validation and conditional replacements where appropriate.
- Standardized data types for dates, numerical measures, and categorical fields.

## Exploratory Financial Analysis

Exploratory analysis was conducted using Excel PivotTables to understand revenue growth patterns, profitability distribution, and key performance differences across products, customer segments, regions, and discount levels. The objective of this phase was to surface patterns, concentration risks, and anomalies prior to deeper profitability diagnostics and forecasting.

### Data Context

The dataset spans the final four months of 2013 and the full 2014 financial year. Due to the partial nature of 2013 data, 2014 was treated as the primary analytical year for trend and seasonality assessment to ensure comparability and structural completeness.

### Revenue & Profit Trend Analysis

Revenue increased significantly between 2013 and 2014; however, because 2013 includes only Q4 data, full year-over-year comparison would overstate organic growth. For this reason, analysis focused on 2014 monthly trends.

Within 2014, revenue exhibited clear seasonality:

- Revenue peaked in **October 2014**
- Declined in **November 2014**
- Rebounded in **December 2014**

Profit generally followed revenue patterns, although November 2014 recorded a **25.91% year-over-year decline** compared to November 2013, suggesting temporary margin compression during that period.

Overall, 2014 demonstrates sustained upward momentum with identifiable seasonal fluctuations concentrated in Q4.

**Product Profitability**

Paseo is the strongest-performing product, generating **$4,797,437.95 in total profit**, followed by VTT. Carretera ranks lowest in absolute profit contribution.

Profit contribution is moderately distributed across the product portfolio, indicating diversification rather than excessive dependency on a single product line. However, performance disparity suggests portfolio optimization opportunities.

**Segment Performance**

The Government segment dominates profitability, contributing **67.41% of total profit**, indicating heavy reliance on this segment for financial performance.

The Enterprise segment contributes **–3.64%**, making it loss-making and dilutive to overall margin performance.

This concentration profile suggests strategic dependency risk and potential need for Enterprise pricing or cost structure reassessment.

**Regional Contribution**

Profit distribution across the five operating regions appears relatively balanced. France leads in total profit contribution, followed by Germany, while Mexico contributes the least.

Although regional dominance is not extreme, margin comparison indicates that revenue concentration does not always align proportionally with profitability strength, warranting closer regional margin management.

**Discount Impact on Margins**

Discount band analysis indicates that higher discounting does not proportionally increase profitability:

- Low discount band contributes **36.63% of total profit**
- High discount band contributes **20.06%**
- No-discount transactions contribute **10.28%**

These findings suggest that moderate discounting may stimulate profitable volume, whereas aggressive discounting compresses margins without delivering equivalent profit expansion. This raises potential pricing discipline concerns.

## Profitability Diagnostics

### High-Margin vs Low-Margin Products

Products were categorized into margin bands:

- Negative Margin (<0%)
- Low Margin (0–10%)
- Moderate Margin (10–20%)
- High Margin (>20%)

Analysis reveals that profitability is concentrated within select high-margin products, while several products operate below the company's average margin threshold.

This indicates potential margin dilution within specific product lines and highlights candidates for pricing review or cost restructuring.

### Loss-Making Discount Bands

Discount band diagnostics show margin compression as discount levels increase. While not all discount tiers are loss-making, margin erosion becomes evident beyond moderate discount thresholds.

This suggests diminishing profitability returns at higher discount rates and supports implementing structured discount governance.

### Segment Margin Differences

Channel Partners demonstrates a strong margin profile of **73.13%**, reflecting efficient revenue-to-profit conversion.

In contrast, the Small Business segment records a **9.77% margin**, significantly below the company average.

The disparity suggests structural differences in pricing power, cost-to-serve, or discounting behavior across segments.

Enterprise remains structurally unprofitable, reinforcing the need for strategic review.

**Volume vs Profitability Tradeoffs**

Volume vs profitability analysis examined Units Sold alongside Profit and Margin.

Findings indicate that high-volume products generally align with strong profitability performance, suggesting that growth has not been margin-dilutive overall.

However, certain high-volume SKUs operate below the average margin, indicating selective volume-driven margin compression.

This distinction is critical: growth quality varies across the portfolio.

## Forecasting Analysis

**Methodology**

Revenue forecasting was conducted using:

- PivotTable-based monthly aggregation
- FORECAST.ETS() formula
- Excel Forecast Sheet

Both forecasting methods produced consistent seasonal detection and trend projection results.

2014 monthly data was used as the primary baseline due to its completeness and full seasonal cycle representation.

**Seasonality Assessment**

Historical 2014 data exhibits identifiable seasonal behavior:

- Q4 peak in October
- November dip
- December rebound

Both the ETS formula and Forecast Sheet detected this seasonal structure within the historical data.

**Forecast Projection (6 Months)**

The 6-month projection indicates:

- Continued upward revenue trajectory
- Smoother seasonal fluctuations compared to historical volatility

- Moderate volatility within confidence intervals

Because the dataset includes only one complete annual cycle, the forecasting model prioritizes established growth trend continuation over aggressive seasonal repetition.

This suggests sustained revenue momentum, though long-term cyclicality cannot be robustly confirmed with the current historical span.

**Analytical Integrity Note**

Due to partial 2013 data (Q4 only), year-over-year full-year comparisons were adjusted to avoid overstating growth. Forecast interpretation acknowledges limited multi-year seasonal repetition.

**Executive-Level Summary**

Kraven demonstrates strong revenue growth and concentrated profitability within the Government segment and select high-margin products. Discounting strategy appears moderately effective at lower tiers but erodes margins at higher thresholds. Forecast analysis projects continued upward momentum, supported by stable demand patterns and observable seasonality.

However, dependency on a single segment and structural underperformance in Enterprise represent financial risk areas requiring strategic attention.

## Dashboard Overview —Financial Performance Analysis

**Page 1 — Revenue Performance Overview**

**Objective:**
To provide a clear view of Kraven's revenue generation, product contribution, and segment performance during 2014, highlighting trends, peaks, and regional distribution.

**Key Features & Visuals:**

1. **KPI Header Section:**
    - Revenue: $118.73M
    - Profits: $16.89M
    - Profit Margin: 14.23%
    - Units Sold: 1.13M
2. **Revenue Trends:**
    - Monthly line chart showing clear seasonal fluctuations.
    - Peak in October 2014 and a dip in November 2014, with recovery in December.
3. **Product Contribution:**
    - Bar chart highlighting top-performing products: Paseo, VTT, and Amarilla.
    - Low performers like Carretera are highlighted in red to flag underperforming SKUs.
4. **Regional Performance:**

    - Revenue contribution by country visualized as a bar chart.
    - France and Germany lead performance, while Mexico contributes the least.
5. **Segment Performance:**
    - Segment profitability visual shows Government dominance (67.41%), with Enterprise negative contribution highlighted.
    - Supports identification of high- and low-value segments.

**Narrative:**
This page provides a comprehensive view of where revenue is coming from, the contribution of products and segments, and regional performance trends. It lays the foundation for deeper profitability and risk analysis.

**Page 2 — Profit Sustainability & Margin Risk Assessment**

**Objective:**
To evaluate the efficiency, sustainability, and risks of Kraven's profit structure, focusing on margin quality, discount impact, and segment dependency.

**Key Features & Visuals:**

1. **Top KPI Section (Risk Indicators):**
    - % Profit from Government Segment: 67.41%
    - Loss-Making Segment: Enterprise –3.1%
    - Lowest Product Margin: Velo 12.6%
    - Forecasted 6-Month Growth: –2.6%
2. **Margin Quality Across Segments:**
    - Bar chart sorted descending by weighted profit margin.
    - Provides insight into segment efficiency (Enterprise: loss, Small Business: thin margin, Channel Partners: high efficiency).
3. **Margin Performance by Product:**
    - Highlights high- and low-margin products.
    - Colors indicate margin risk.
4. **Margin Compression Across Discount Levels:**
    - Donut chart visualizes how high discount bands reduce margins.
    - Supports pricing and discount policy decisions.
5. **Revenue Forecast & Trend:**
    - Line chart showing historical revenue (2014) and projected 6-month trend.
    - Confidence intervals illustrate potential variability.
    - Forecast indicates a modest –2.6% normalization from peak December revenue, reflecting seasonal adjustment.

**Narrative:**
This page provides actionable insight into margin drivers, discount effects, and segment risk exposure. The combination of KPIs, margin charts, and forecast signals supports data-driven decisions to improve sustainability and reduce dependency risk.

## Business Insights & Recommendations

**Expected Outcome:**
Leadership expects this project to:

- Clarify true profitability drivers
- Identify margin risks related to discounting
- Highlight high- and low-value products and segments
- Validate financial figures through recalculation
- Provide a forward-looking revenue outlook
- Support data-driven pricing and planning decisions

**Narrative for Documentation:**
The dashboard provides leadership with a clear, executive-ready view of Kraven's financial performance. Page 1 identifies where revenue originates, top and bottom products, and segment contributions. Page 2 evaluates sustainability by analyzing segment efficiency, margin compression from discounts, and projected revenue trends. Together, the two pages allow for immediate identification of risk areas and opportunities, supporting actionable decision-making.